Chapter 1 Classical Algebra

Exercise 5

Let $S = \{p + q\alpha + r\alpha^2 \mid p, q, r \in \mathbb{Q}\}$. We first prove that S is a subring of $\mathbb{C}$.

Note that $1 = 1+0\alpha+0\alpha^2 \in S$. Let $p+q\alpha+r\alpha^2, p'+q'\alpha+r'\alpha^2 \in S$, where $p, q, r, p', q', r' \in \mathbb{Q}$. Then we have $(p+q\alpha+r\alpha^2)(p'+q'\alpha+r'\alpha^2) = (p+p')+(q+q')\alpha+(r+r')\alpha^2 \in S$, $-(p+q\alpha+r\alpha^2) = (-p)+(-q)\alpha+(-r)\alpha^2 \in S$ and $(p+q\alpha+r\alpha^2)(p'+q'\alpha+r'\alpha^2) = (pp'+2rq')+(pq'+p'q+2rr')\alpha+(pr'+qq'+qr'+rp')\alpha^2 \in S$. By the subring test, S is a ring.

Next, we construct an inverse explicitly to show that S is a subfield. Let $x = p+q\alpha+r\alpha^2 \in S\setminus\{0\}$, where $p, q, r \in \mathbb{Q}$. Then we have $x^2 = (p^2+4qr)+(2pq+2r^2)\alpha+(2pr+q^2)\alpha^2$ and $x^3 = (p^3+12pqr+2q^3+4r^3)+(3p^2q+6pr^2+6q^2r)\alpha+(3p^2r+3pq^2+6qr^2)\alpha^2$. Let $x^3+k_2x^2+k_1x+k_0 = 0$. Denote the coefficient of α^j in x^i by x_{ij}. Then we have $(x_{30}+x_{31}\alpha+x_{32}\alpha^2)+k_2(x_{20}+x_{21}\alpha+x_{22}\alpha^2)+k_1(p+q\alpha+r\alpha^2)+k_0 = 0$, that is, $(x_{30} + x_{20}k_2 + pk_1 + k_0) + (x_{31} + x_{21}k_2 + qk_1)\alpha + (x_{32} + x_{22}k_2 + rk_1)\alpha^2 = 0$. Since $1, \alpha, \alpha^2$ are linearly independent, comparing coefficients gives us the linear system

$$\begin{cases} k_0 + pk_1 + x_{20}k_2 = -x_{30} \\ qk_1 + x_{21}k_2 = -x_{31} \\ rk_1 + x_{22}k_2 = -x_{32} \end{cases}.$$

We can use the following python code to solve this system.

```python
import sympy as sp

k0, k1, k2 = sp.symbols('k0 k1 k2')
p, q, r = sp.symbols('p q r')

x30 = p**3 + 12 * p * q * r + 2 * (q**3) + 4 * (r**3)
x31 = 3 * (p**2) * q + 6 * p * (r**2) + 6 * (q**2) * r
x32 = 3 * (p**2) * r + 3 * p * (q**2) + 6 * q * (r**2)
x20 = p**2 + 4 * q * r
x21 = 2 * p * q + 2 * (r**2)
x22 = 2 * p * r + q**2

M = sp.Matrix([
    [1, p, x20, - x30],
    [0, q, x21, - x31],
    [0, r, x22, - x32]
])

solution = sp.solve_linear_system(M, k0, k1, k2)

for var in [k0, k1, k2]:
    print(f"{var} =")
    sp.pprint(sp.simplify(solution[var]))
```

The solution is $k_0 = -p^3+6pqr-2q^3-4r^3$, $k_1 = 3p^2-6qr$ and $k_2 = -3p$. Since $x^3+k_2x^2+k_1x+k_0 = 0$, we have $x(x^2+k_2x+k_1+k_0x^{-1}) = 0$. Recall that $\mathbb{C}$ is a field and hence a domain, and $x \neq 0$. Therefore, $x^2 + k_2x + k_1 + k_0x^{-1} = 0$, that is,

$$x^{-1} = -\frac{1}{k_0}x^2 - \frac{k_2}{k_0}x - \frac{k_1}{k_0}.$$

Since x and x^2 are linear combinations of $1, \alpha, \alpha^2$, x^{-1} is also a linear combination of $1, \alpha, \alpha^2$. $\square$